Evaluation of Generated Enzymes from Pretrained ProteinNet

	ESP score				AlphaFold2 pLDDT				% Under 2Å RMSD			
	①	②	③	④	①	②	③	④	①	②	③	④
DNA-directed DNA polymerase	0.52	0.35	0.67	0.73	72.2	74.2	85.5	82.0	58.1	21.5	64.5	63.4
lysozyme	0.29	0.42	0.21	0.25	79.1	77.3	84.7	91.2	17.9	71.4	78.6	82.1
β-lactamase	0.86	0.85	0.87	0.91	76.6	78.4	90.7	80.2	50.0	60.7	100.0	89.3
endothiapepsin	0.27	0.32	0.16	0.31	78.7	78.3	87.0	93.9	80.0	60.0	80.0	96.0
RNA helicase	0.83	0.88	0.72	0.90	77.5	69.7	88.6	75.0	19.0	71.4	66.7	52.4
tRNA-guanosine preQ1 transglycosylase	0.16	0.22	0.17	0.15	75.8	75.7	80.3	77.7	50.0	100.0	64.3	78.6
micrococcal nuclease	0.90	0.96	0.91	0.88	71.0	88.1	84.6	91.9	44.4	0.0	100.0	100.0
ribonuclease T1	0.89	0.97	0.98	0.99	67.6	66.2	91.9	98.1	100.0	62.5	100.0	100.0
SARS-COV-2 M^{pro}	0.89	0.90	0.94	0.98	72.0	76.4	65.8	89.0	0.0	50.0	62.5	100.0
alcohol dehydrogenase	0.28	0.57	0.77	0.48	70.9	68.3	91.2	71.9	75.0	12.5	87.5	50.0
Average	0.59	0.64	0.64	0.66	74.1	75.3	85.0	85.1	49.4	51.0	80.4	81.2

① Inpainting ② RFDiffusion + ProteinMPNN ③ Alphafold3 + LigandMPNN ④ ProteinNet